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Advancements of Large Language Models for Enhancing Carbon Capture Technologies: A Comprehensive Review

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Abstract:
This paper reviews the current research status, challenges, and prospects of applying large language models (LLMs) in carbon capture technologies. The review emphasizes the importance of interdisciplinary research, integrating AI into chemistry, engineering, and environmental science to address complex challenges in carbon capture. It provides a detailed analysis of how LLMs can be utilized across various stages of carbon capture, from experimental design to industry implementation, showcasing their potential to accelerate innovation. It also reveals the use of LLMs to support gathering and analyzing sustainable information, such as carbon tax, carbon footprint, and social analysis. LLMs not only show great potential in designing and discovering materials for carbon capture technologies but also are promising to accelerate the whole industry's development through their powerful data processing and pattern recognition capabilities. In addition, the review paper also discusses challenges in the application of LLMs for carbon capture technologies and future directions and prospects.

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Submitted By: Name: AKKINAPELLY PRATHIBA

Roll Number: 1602-23-748-035

Class: 3/4-B.E-CSE-AIML

Department: Department of Computer Science and Engineering

College: VASAVI COLLEGE OF ENGINEERING